

Feature	Contrast	Effect	Simultaneous 95% CI	Decision
helpful mass share	Muon–AdamW	-0.0018016	[-0.0095111, +0.00590791]	inconclusive
helpful mass share	NorMuon–AdamW	+0.000437186	[-0.00522206, +0.00609643]	inconclusive
helpful mass share	NorMuon–Muon	+0.00223878	[-0.00674053, +0.0112181]	inconclusive
helpful participation	Muon–AdamW	+0.00358455	[+0.000143587, +0.00702552]	positive
helpful participation	NorMuon–AdamW	+0.00235085	[-0.00269661, +0.0073983]	inconclusive
helpful participation	NorMuon–Muon	-0.00123371	[-0.00668711, +0.0042197]	inconclusive
degrading mass	Muon–AdamW	-0.00130579	[-0.00396615, +0.00135456]	inconclusive
degrading mass	NorMuon–AdamW	-0.00183596	[-0.00400392, +0.00033199]	inconclusive
degrading mass	NorMuon–Muon	-0.000530169	[-0.00286867, +0.00180833]	inconclusive

Table 1: All nine final-stage co-primary contrasts, averaging every declared rate. One common-resample max- T family covers all entries; task resampling does not estimate training-seed variation.

Feature	RMSE reduction	Improved	Defined	Criterion
helpful mass share	-0.000389109	2/4	4/4	not supported
helpful participation	-0.00434356	1/4	4/4	not supported
degrading mass	+0.0100691	4/4	4/4	supported
50% random-removal retention	+0.000341663	2/4	4/4	not supported

Table 2: All four functional predictors beyond optimizer, stage and within-optimizer dose. Decisions use exact MSE comparisons; displayed RMSE is not a significance test. Undefined folds are retained, never pooled as available cases.

Coordinate utility and held-dose retrieval

Complete corrected DenseOn matrix: 61 states, 224 fixed queries, 14 tasks. This readout reports the fixed measurements and comparisons; the complete NAACL manuscript is a separate deliverable.

Muon does not meet the joint native-coordinate criterion relative to AdamW; this does not establish equivalence. The required mean directions do not hold under every tested shared rotation. NorMuon does not meet the joint native-coordinate criterion relative to AdamW; this does not establish equivalence. The required mean directions do not hold under every tested shared rotation. With 50% of coordinates removed, relative shortlist nDCG is 0.991638/0.993778/0.992056 for AdamW/Muon/NorMuon. The figure reports all declared rates and stages.

Helpful mass share does not meet the declared held-out-dose predictive criterion. Helpful participation does not meet the declared held-out-dose predictive criterion. Degrading mass meets the declared held-out-dose predictive criterion. 50% random-removal retention does not meet the declared held-out-dose predictive criterion. This tests prediction outside a learning-rate dose on the same task suite, not causal mediation or generalization to unseen tasks.

Coordinate-level utility, its tested-basis sensitivity, and held-out-dose prediction are distinct results. Shared-rotation sign stability does not establish arbitrary-basis invariance or semantic independence of dimensions; these measurements alone do not identify a mediator of an optimizer effect.

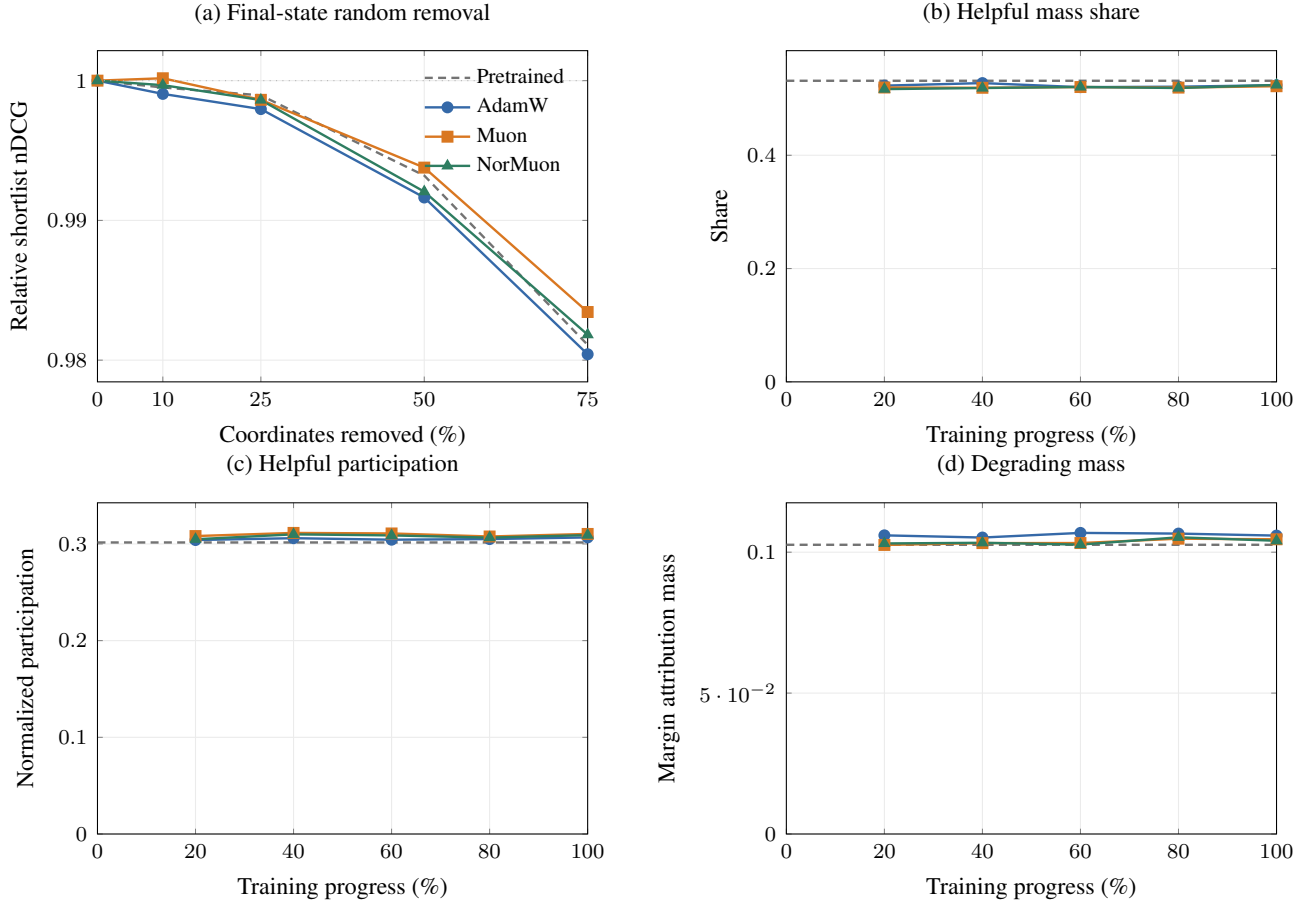


Figure 1: Dimension utility over the complete rate grid. (a) Final-state retention after shared random coordinate removal; the zero-removal value is one by definition. (b–d) Native-coordinate margin attribution at all five stages. Each optimizer curve averages all four rates and 14 tasks; (a) additionally averages 20 shared masks per nonzero fraction. The gray curve or horizontal line is the pretrained reference. Lines join measured points, not fitted trajectories. These are descriptive means, not confidence intervals or full-corpus retrieval scores; formal contrasts and rotation controls are reported separately.